

DATA ANALYSIS · SQL · PYTHON · TABLEAU

fashion retail intelligence

Merchandising, customer and demand analytics on 31.8 million H&M transactions, September 2018 to September 2020.

THE BRIEF

three questions a merchandising team asks every season

31.8M

transactions

1.36M

customers

105K

articles

2 yrs

Sep 2018 – Sep 2020

1. **Merchandising:** which categories drive sales, and how hard does the range work?
2. **Customers:** who creates the value, and do new customers come back?
3. **Demand:** how seasonal is it, and can it be forecast a quarter ahead?

SQL · DuckDB

Python · pandas

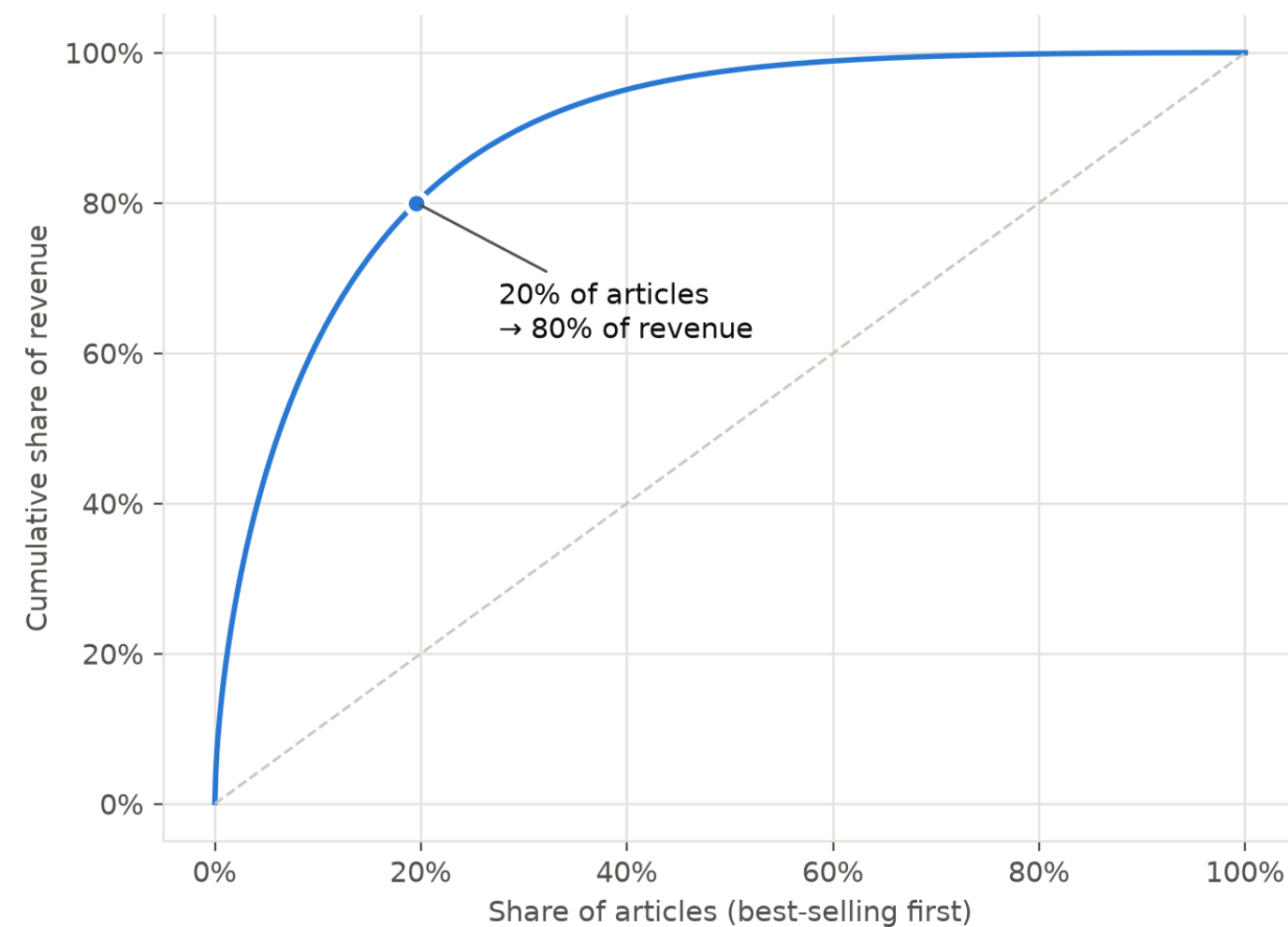
scikit-learn

Tableau Public

ASSORTMENT CONCENTRATION

a fifth of the range drives 80% of revenue

A small share of articles drives most revenue



19.5%

of articles generate 80% of revenue

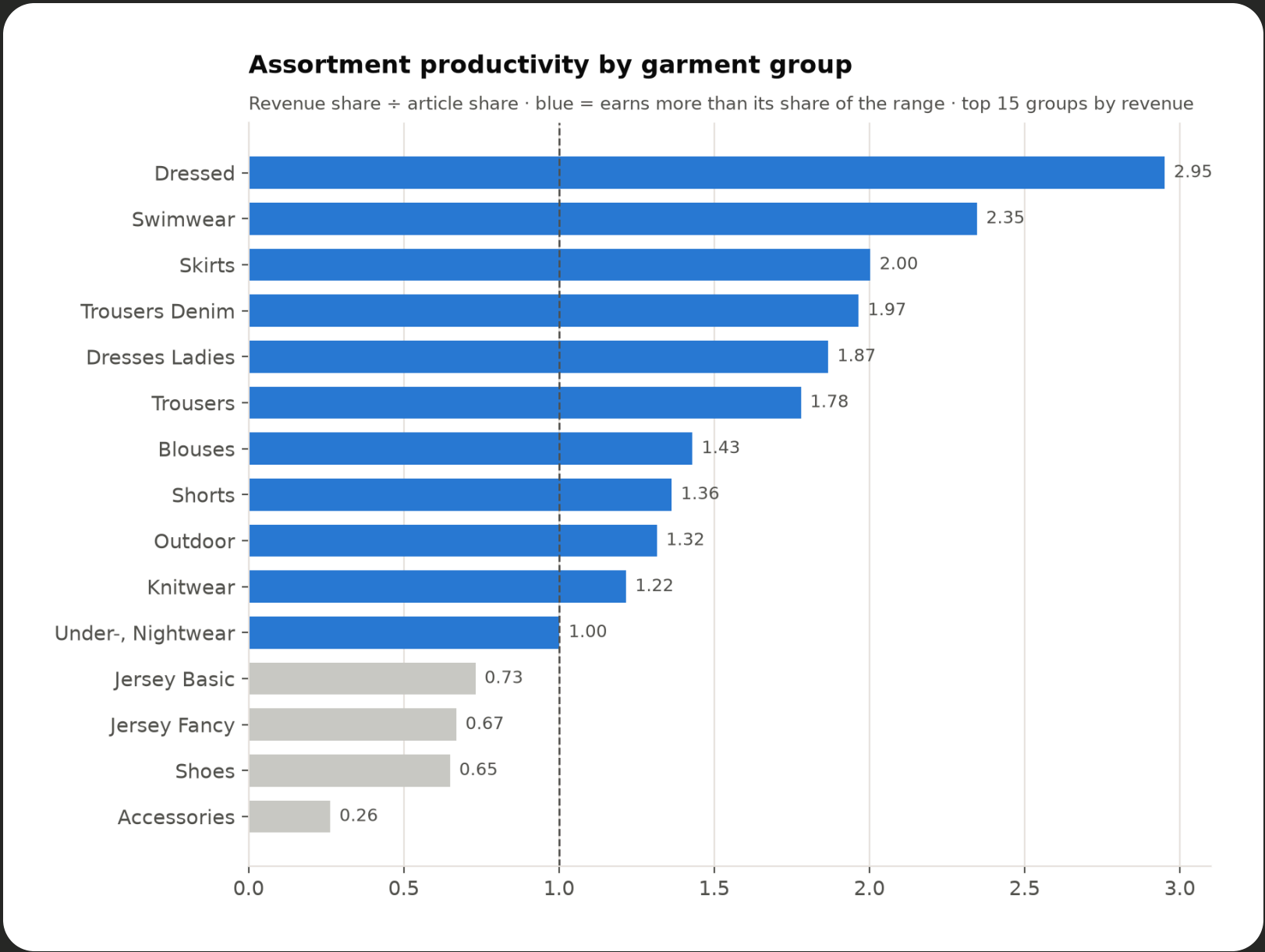
2.4%

of revenue comes from the bottom half of the range

Range decisions at the top of the curve matter far more than those in the long tail.

PRODUCTIVITY AND MARKDOWNS

some categories punch far above their weight



2.95×

Dressed earns nearly 3× its share of the range

0.26×

Accessories: 11% of articles, 3% of revenue

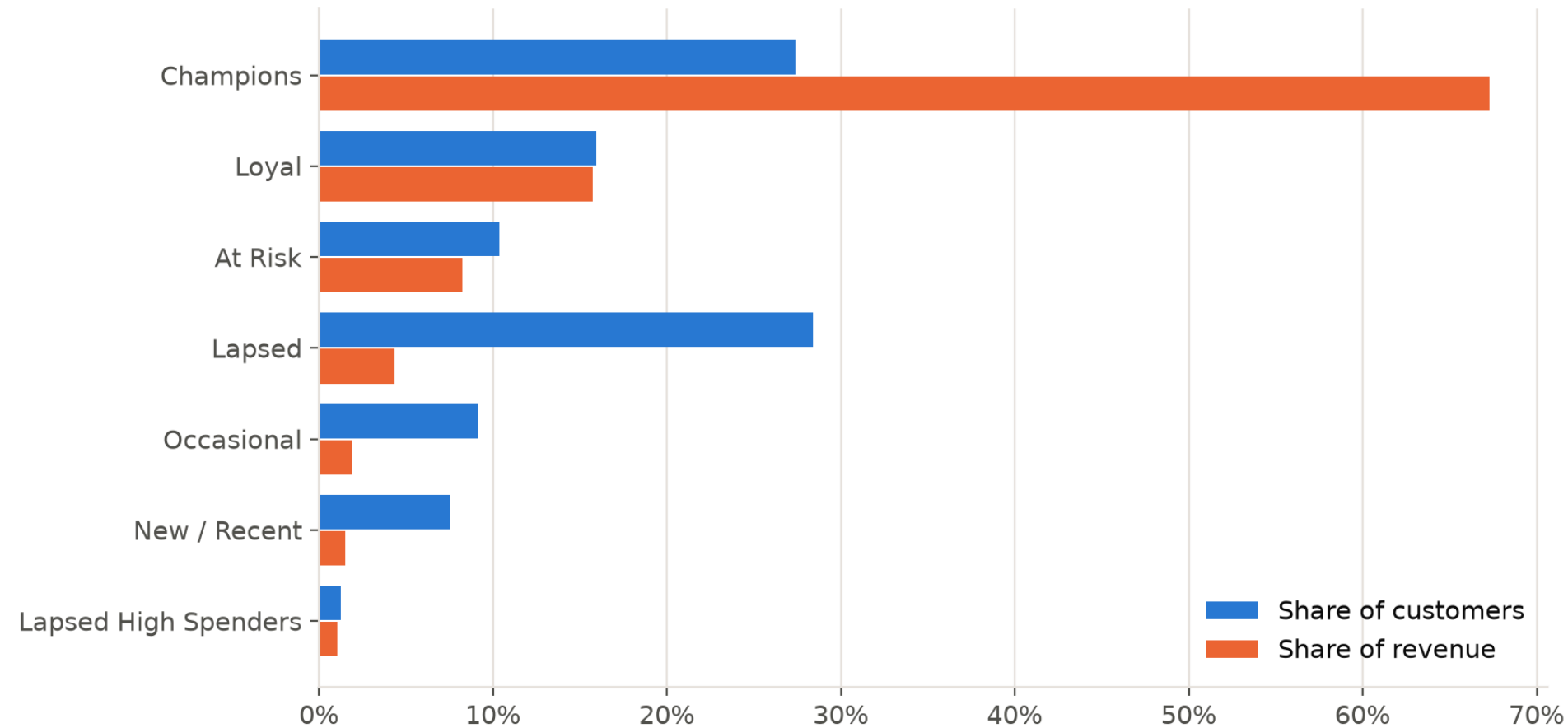
33%

of units sell on markdown; 46% in Socks and Tights

CUSTOMER VALUE

value sits with loyal, omnichannel shoppers

Customer segments: size vs. value



67%

of revenue from Champions, 27% of customers

68%

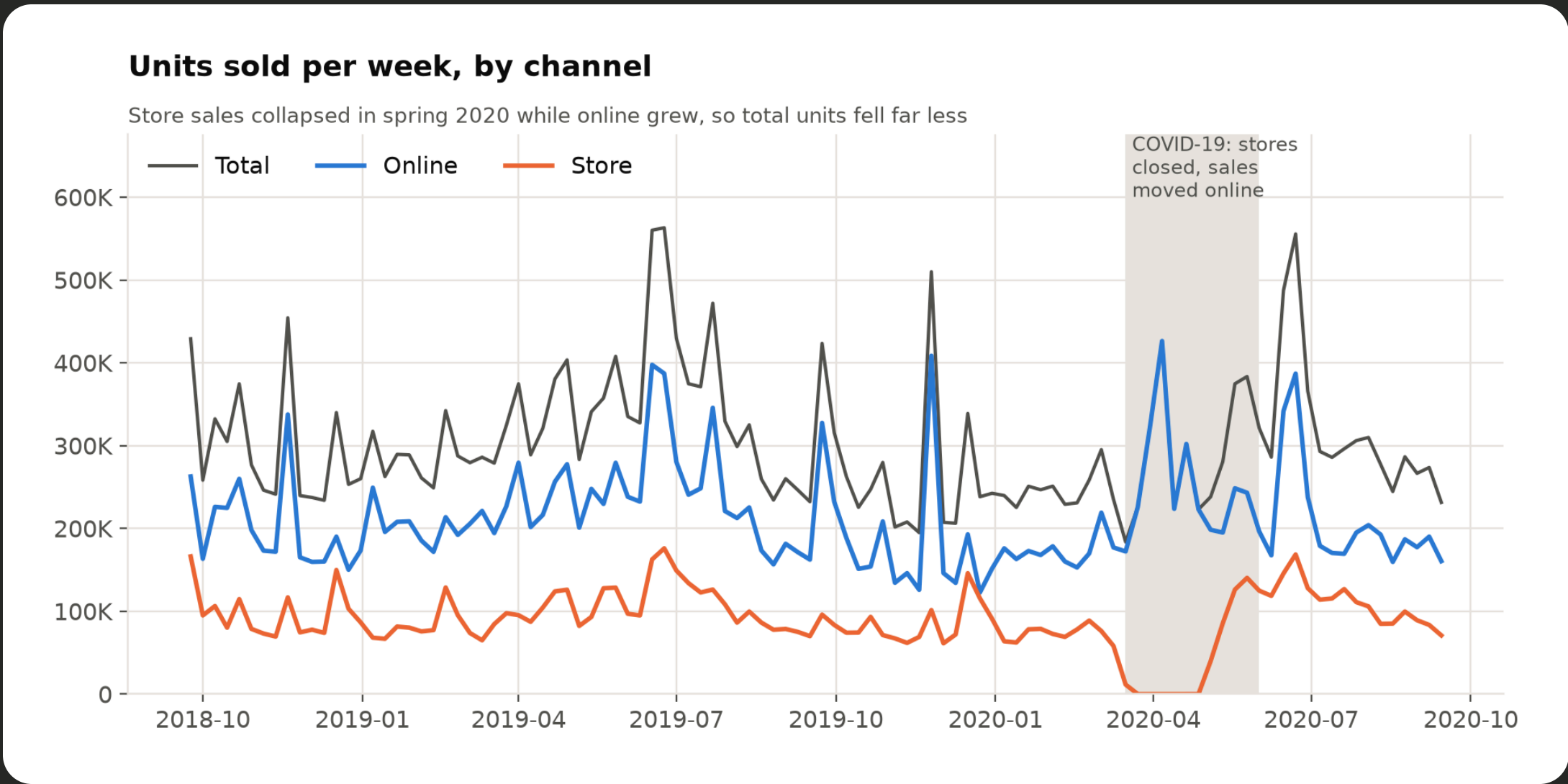
from omnichannel shoppers, 36% of customers

37.5%

of new customers return within 90 days

CHANNEL SHIFT

stores closed, and online absorbed much of the drop



-65%

store units

+7%

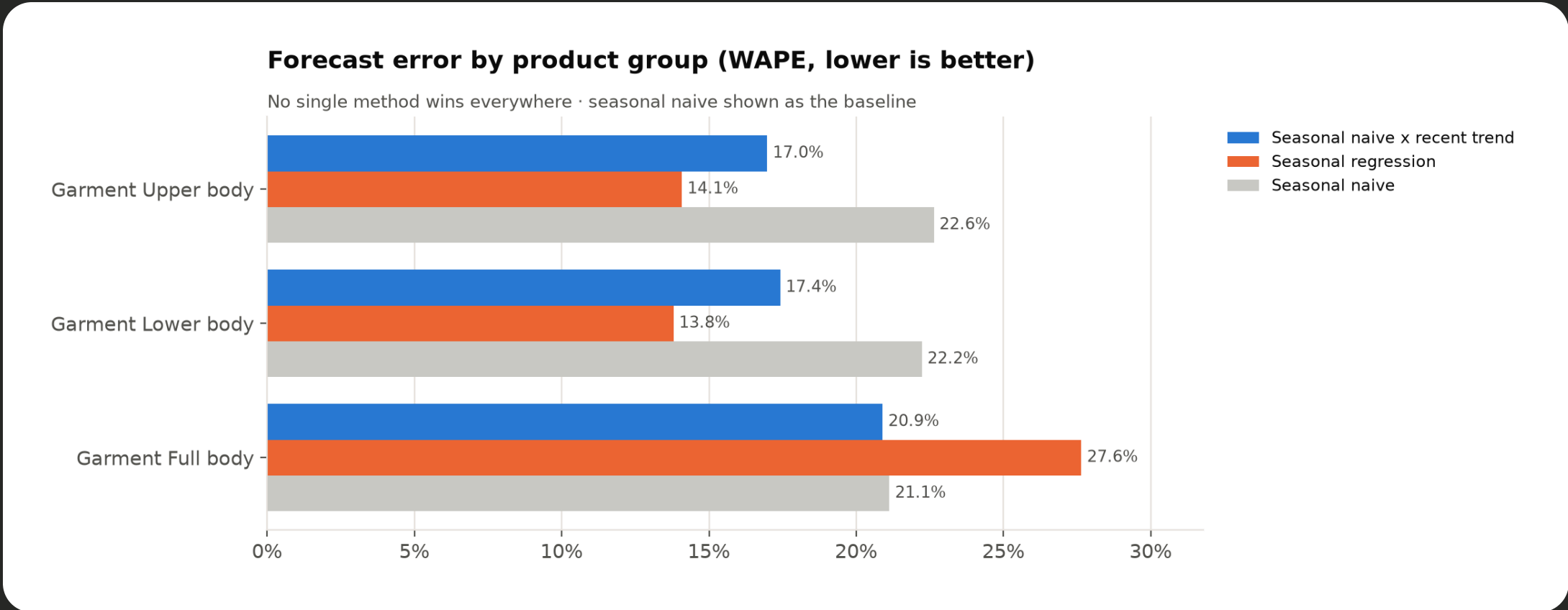
online units

-15%

total units

DEMAND FORECASTING

no single forecast method wins everywhere



13.8%

WAPE, lower body: regression wins

20.9%

WAPE, full body: naive × trend wins

Averages tie at 18.4% vs. 18.5%, which hides the gap. Pick the method per product group.

RECOMMENDATIONS

four next steps for the business

01

Trim the long tail

Review low sellers in low-productivity groups like Accessories; protect Dressed, Swimwear, Skirts and Denim.

02

Probe markdown-heavy groups

Where 40%+ of units sell on markdown, compare buy depth, pricing and timing with actual demand.

03

Retain the top tier

Protect Champions and omnichannel shoppers; test win-back offers for lapsed customers against a holdout.

04

Forecast by product group

Back-test methods for each group every planning cycle and use the best one, not one model for all.

Starting points, not final decisions: the data has no inventory or margin information, and the findings are descriptive.